

MCQs from Lab 01 and Lab 03 Manuals

Below are 80+ MCQs. The correct option is bolded and marked with .

Lab 01 — Computer Vision Basics

1. Computer vision primarily aims to:

- A) Store images in databases
- B) Enable computers to interpret and understand visual information
- C) Compress images for the internet
- D) Convert analog signals to digital only

 Correct Answer: B) Enable computers to interpret and understand visual information

2. Which of the following is NOT a basic terminology in computer vision?

- A) Pixel
- B) Resolution
- C) Grayscale
- D) Compiler

 Correct Answer: D) Compiler

3. A pixel is best defined as:

- A) The largest unit of an image
- B) The smallest unit of an image carrying color/intensity
- C) A type of image filter
- D) A compression algorithm

 Correct Answer: B) The smallest unit of an image carrying color/intensity

4. Resolution of an image is typically expressed as:

- A) File size in MB
- B) Dimensions in pixels, e.g., 1920×1080
- C) Number of colors
- D) Compression ratio

 Correct Answer: B) Dimensions in pixels, e.g., 1920×1080

5. In a grayscale image, pixel values typically range from:

- A) 0 to 100
- B) 0 to 255
- C) 1 to 10
- D) -255 to 255

 Correct Answer: B) 0 to 255

6. RGB stands for:

- A) Red, Gray, Blue
- B) Red, Green, Blue
- C) Red, Green, Black
- D) Ratio, Green, Blue

✓ Correct Answer: B) Red, Green, Blue

7. Feature extraction is:

- A) Removing all features from an image
- B) Identifying and isolating relevant features from an image
- C) Compressing an image
- D) Rotating an image

✓ Correct Answer: B) Identifying and isolating relevant features from an image

8. Edge detection is used to:

- A) Fill colors in an image
- B) Identify boundaries or transitions between regions
- C) Increase image resolution
- D) Convert RGB to grayscale

✓ Correct Answer: B) Identify boundaries or transitions between regions

9. Object detection means:

- A) Identifying and localizing specific objects in an image/video
- B) Only classifying an image as cat or dog
- C) Removing background
- D) Rotating objects

✓ Correct Answer: A) Identifying and localizing specific objects in an image/video

10. Segmentation is:

- A) Dividing an image into meaningful segments or regions
- B) Increasing brightness
- C) Adding text to an image
- D) Blurring an image

✓ Correct Answer: A) Dividing an image into meaningful segments or regions

11. A Convolutional Neural Network (CNN) is:

- A) A type of database
- B) A deep learning model designed for grid-structured data like images
- C) A compression algorithm
- D) An image file format

✓ Correct Answer: B) A deep learning model designed for grid-structured data like images

12. Deep learning is:

- A) A subset of machine learning using neural networks with multiple layers

- B) A type of image resizing
- C) A thresholding technique
- D) A color conversion method

✓ Correct Answer: A) A subset of machine learning using neural networks with multiple layers

13. A feature map is:

- A) A map of image coordinates
- B) Output generated by applying convolutional filters to an image
- C) A type of histogram
- D) A grayscale image

✓ Correct Answer: B) Output generated by applying convolutional filters to an image

14. Classification assigns:

- A) A label or category to an input image
- B) A new coordinate system
- C) A blur kernel
- D) A compression ratio

✓ Correct Answer: A) A label or category to an input image

15. Image registration is:

- A) Aligning multiple images to a common coordinate system
- B) Registering an image in a database
- C) Adding text to an image
- D) Converting to grayscale

✓ Correct Answer: A) Aligning multiple images to a common coordinate system

16. Optical flow estimates:

- A) Motion of objects between consecutive video frames
- B) Color distribution
- C) Image resolution
- D) Compression rate

✓ Correct Answer: A) Motion of objects between consecutive video frames

17. Tracking means:

- A) Following movement of objects across multiple frames
- B) Detecting edges
- C) Enhancing contrast
- D) Cropping an image

✓ Correct Answer: A) Following movement of objects across multiple frames

18. Homography is:

- A) A transformation relating two images of the same planar surface
- B) A type of blur

C) A histogram equalization method

D) A color space

☒ Correct Answer: A) A transformation relating two images of the same planar surface

19. A feature descriptor is:

A) A compact representation of a feature for matching/recognition

B) A large raw pixel array

C) A type of image file

D) A blur kernel

☒ Correct Answer: A) A compact representation of a feature for matching/recognition

20. A histogram represents:

A) Distribution of pixel intensities in an image

B) Image resolution

C) Number of channels

D) Compression ratio

☒ Correct Answer: A) Distribution of pixel intensities in an image

21. SIFT stands for:

A) Scale-Invariant Feature Transform

B) Simple Image Filtering Technique

C) Scaled Intensity Frequency Transform

D) Standard Image Format Type

☒ Correct Answer: A) Scale-Invariant Feature Transform

22. SURF is similar to SIFT but:

A) Slower

B) Faster

C) Only for grayscale

D) Only for video

☒ Correct Answer: B) Faster

23. HOG stands for:

A) Histogram of Oriented Gradients

B) High-Order Grayscale

C) Horizontal Orientation Grid

D) Histogram of RGB

☒ Correct Answer: A) Histogram of Oriented Gradients

24. Which is an application of computer vision?

A) Autonomous vehicles

B) Medical imaging

C) Facial recognition

D) All of the above

✓ Correct Answer: D) All of the above

25. Which library is mainly used for general computer vision tasks?

- A) OpenCV
- B) Keras
- C) Caffe
- D) Detectron2

✓ Correct Answer: A) OpenCV

26. TorchVision is primarily designed for:

- A) PyTorch
- B) TensorFlow
- C) Caffe
- D) OpenVINO

✓ Correct Answer: A) PyTorch

27. Detectron2 is specialized in:

- A) Object detection
- B) NLP
- C) Image compression
- D) Audio processing

✓ Correct Answer: A) Object detection

28. Hugging Face is known for:

- A) Pretrained NLP and CV models
- B) Only image blurring
- C) Only thresholding
- D) Only image rotation

✓ Correct Answer: A) Pretrained NLP and CV models

29. A digital image is defined as a function:

- A) $f(x, y)$
- B) $g(a, b, c)$
- C) $h(t)$
- D) $p(q)$

✓ Correct Answer: A) $f(x, y)$

30. In image coordinates, the origin is usually at:

- A) Bottom-left
- B) Top-left
- C) Center
- D) Top-right

✓ Correct Answer: B) Top-left

31. In image coordinates, the x-axis goes:

- A) Right
- B) Left
- C) Up
- D) Down

✓ Correct Answer: A) Right

32. In image coordinates, the y-axis goes:

- A) Right
- B) Left
- C) Up
- D) Down

✓ Correct Answer: D) Down

33. In digital image representation, M represents:

- A) Total rows / height
- B) Total columns / width
- C) Number of channels
- D) Intensity

✓ Correct Answer: A) Total rows / height

34. In digital image representation, N represents:

- A) Total rows / height
- B) Total columns / width
- C) Number of channels
- D) Intensity

✓ Correct Answer: B) Total columns / width

35. The first fundamental step of Digital Image Processing is:

- A) Image Enhancement
- B) Image Acquisition
- C) Segmentation
- D) Compression

✓ Correct Answer: B) Image Acquisition

36. Image enhancement is used to:

- A) Highlight interesting features like brightness and contrast
- B) Compress the image
- C) Segment the image
- D) Recognize objects

✓ Correct Answer: A) Highlight interesting features like brightness and contrast

37. Image restoration is used to:

- A) Improve the appearance of an image

- B) Reduce image resolution
- C) Convert to binary
- D) Add noise

☒ Correct Answer: A) Improve the appearance of an image

38. Wavelets and multi-resolution processing are used for:

- A) Data compression and pyramidal representation
- B) Color conversion
- C) Object detection
- D) Text addition

☒ Correct Answer: A) Data compression and pyramidal representation

39. Compression is important because:

- A) It reduces storage requirement
- B) It increases resolution
- C) It adds color
- D) It detects edges

☒ Correct Answer: A) It reduces storage requirement

40. Morphological processing deals with:

- A) Extracting image components useful for shape representation
- B) Histogram equalization
- C) Color space conversion
- D) Image rotation

☒ Correct Answer: A) Extracting image components useful for shape representation

41. Which DIP step is considered the most difficult?

- A) Image Acquisition
- B) Segmentation
- C) Compression
- D) Image Enhancement

☒ Correct Answer: B) Segmentation

42. Representation and description follow:

- A) Segmentation
- B) Compression
- C) Acquisition
- D) Enhancement

☒ Correct Answer: A) Segmentation

43. Object recognition assigns:

- A) Labels to objects based on descriptors
- B) New pixel values

C) Compression ratios

D) Blur kernels

✓ Correct Answer: A) Labels to objects based on descriptors

44. The Knowledge Base in DIP:

A) Is the last stage and limits searching processes

B) Is the first stage

C) Only handles compression

D) Replaces segmentation

✓ Correct Answer: A) Is the last stage and limits searching processes

45. OpenCV loads color images in which format by default?

A) RGB

B) BGR

C) Grayscale

D) HSV

✓ Correct Answer: B) BGR

46. Matplotlib expects images in which format?

A) BGR

B) RGB

C) Grayscale only

D) Binary

✓ Correct Answer: B) RGB

47. Which function converts BGR to RGB in OpenCV?

A) cv2.cvtColor()

B) cv2.imread()

C) cv2.resize()

D) cv2.threshold()

✓ Correct Answer: A) cv2.cvtColor()

48. Grayscale conversion in OpenCV uses:

A) cv2.COLOR_BGR2GRAY

B) cv2.COLOR_RGB2BGR

C) cv2.COLOR_GRAY2BGR

D) cv2.COLOR_BGR2HSV

✓ Correct Answer: A) cv2.COLOR_BGR2GRAY

49. In cv2.resize(), fx and fy represent:

A) Scale factors along x and y

B) Filter types

C) Frame rates

D) Font sizes

✓ Correct Answer: A) Scale factors along x and y

50. In GaussianBlur, the kernel size must be:

- A) Even numbers
- B) Odd numbers
- C) Any integer
- D) Only 3

✓ Correct Answer: B) Odd numbers

51. Cropping an image in OpenCV is done using:

- A) NumPy slicing
- B) cv2.crop()
- C) cv2.cut()
- D) cv2.slice()

✓ Correct Answer: A) NumPy slicing

52. cv2.putText() places text using which reference point?

- A) Bottom-left corner of the text
- B) Top-left corner of the image
- C) Center of the image
- D) Top-right corner of the text

✓ Correct Answer: A) Bottom-left corner of the text

53. In Pandas image analysis, image.reshape(-1, 3) converts:

- A) 3D image array to 2D array
- B) 2D array to 3D array
- C) Grayscale to RGB
- D) RGB to BGR

✓ Correct Answer: A) 3D image array to 2D array

54. cv2.THRESH_BINARY does:

- A) If pixel > threshold, set to max value; else set to 0
- B) If pixel < threshold, set to max value; else set to 0
- C) Keeps all pixels unchanged
- D) Inverts all pixels

✓ Correct Answer: A) If pixel > threshold, set to max value; else set to 0

55. cv2.getRotationMatrix2D() requires:

- A) Center, angle, scale
- B) Width, height, channels
- C) Threshold, max value, type
- D) Kernel size, sigma

✓ Correct Answer: A) Center, angle, scale

56. cv2.warpAffine() uses which matrix size?

- A) 2×2
- B) 2×3
- C) 3×3
- D) 4×4

✓ Correct Answer: B) 2×3

57. cv2.add() uses:

- A) Saturation arithmetic
- B) Modulo arithmetic
- C) Subtraction
- D) Division

✓ Correct Answer: A) Saturation arithmetic

58. Histogram equalization improves:

- A) Contrast
- B) Resolution
- C) File size
- D) Color depth

✓ Correct Answer: A) Contrast

59. cv2.IMREAD_GRAYSCALE loads an image:

- A) Directly as grayscale
- B) As RGB
- C) As BGR
- D) As binary only

✓ Correct Answer: A) Directly as grayscale

Lab 03 — Geometric Transformation

60. A geometric transformation changes:

- A) Pixel color
- B) Spatial position, orientation, scale, or shape
- C) Only brightness
- D) Only contrast

✓ Correct Answer: B) Spatial position, orientation, scale, or shape

61. A photometric transformation changes:

- A) Where the pixel is located
- B) What the pixel looks like (color, brightness, contrast)
- C) Image dimensions
- D) Coordinate system

✓ Correct Answer: B) What the pixel looks like (color, brightness, contrast)

62. Geometric transformation formula is:

- A) $I(x,y) \rightarrow I'(x',y')$
- B) $(x,y) \rightarrow (x',y')$
- C) $f(x,y) \rightarrow g(x,y)$
- D) $p = Ap + b$

☒ Correct Answer: B) $(x,y) \rightarrow (x',y')$

63. Photometric transformation formula is:

- A) $(x,y) \rightarrow (x',y')$
- B) $I(x,y) \rightarrow I'(x',y')$
- C) $A0 = 0$
- D) $p' = Ap$

☒ Correct Answer: B) $I(x,y) \rightarrow I'(x',y')$

64. In image coordinate system, the origin is:

- A) Bottom-left
- B) Top-left
- C) Center
- D) Top-right

☒ Correct Answer: B) Top-left

65. In image coordinate system, x increases:

- A) Right
- B) Left
- C) Up
- D) Down

☒ Correct Answer: A) Right

66. In image coordinate system, y increases:

- A) Right
- B) Left
- C) Up
- D) Down

☒ Correct Answer: D) Down

67. A 2D transformation operates on:

- A) $(x,y) \rightarrow (x',y')$
- B) $(x,y,z) \rightarrow (x',y',z')$
- C) Only 3D points
- D) Only color values

☒ Correct Answer: A) $(x,y) \rightarrow (x',y')$

68. OpenCV's warpAffine() accepts a:

- A) 2×3 matrix
- B) 3×3 matrix
- C) 4×4 matrix
- D) 1×2 matrix

✓ Correct Answer: A) 2×3 matrix

69. OpenCV's warpPerspective() uses a:

- A) 2×2 matrix
- B) 2×3 matrix
- C) 3×3 matrix
- D) 4×4 matrix

✓ Correct Answer: C) 3×3 matrix

70. A 3D homogeneous transformation typically uses a:

- A) 2×3 matrix
- B) 3×3 matrix
- C) 4×4 matrix
- D) 2×2 matrix

✓ Correct Answer: C) 4×4 matrix

71. In a 2×2 transformation matrix, the first column represents:

- A) New landing spot for the X-axis
- B) New landing spot for the Y-axis
- C) Translation in x
- D) Translation in y

✓ Correct Answer: A) New landing spot for the X-axis

72. In a 2×2 transformation matrix, the second column represents:

- A) New landing spot for the X-axis
- B) New landing spot for the Y-axis
- C) Translation in x
- D) Translation in y

✓ Correct Answer: B) New landing spot for the Y-axis

73. In the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, variable 'a' controls:

- A) Horizontal scaling
- B) Vertical shearing
- C) Horizontal shearing
- D) Vertical scaling

✓ Correct Answer: A) Horizontal scaling

74. In the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, variable 'c' controls:

- A) Horizontal scaling

- B) Vertical shearing
- C) Horizontal shearing
- D) Vertical scaling

✓ Correct Answer: B) Vertical shearing

75. In the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, variable 'b' controls:

- A) Horizontal scaling
- B) Vertical shearing
- C) Horizontal shearing
- D) Vertical scaling

✓ Correct Answer: C) Horizontal shearing

76. In the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, variable 'd' controls:

- A) Horizontal scaling
- B) Vertical shearing
- C) Horizontal shearing
- D) Vertical scaling

✓ Correct Answer: D) Vertical scaling

77. An affine transformation preserves:

- A) Collinearity and parallelism
- B) Only color
- C) Only brightness
- D) Only resolution

✓ Correct Answer: A) Collinearity and parallelism

78. The affine transformation equation is:

- A) $p' = Ap$
- B) $p' = Ap + b$
- C) $p' = A + p$
- D) $p' = A/p$

✓ Correct Answer: B) $p' = Ap + b$

79. A linear transformation is an affine transformation:

- A) With translation
- B) Without translation
- C) With perspective
- D) With color change

✓ Correct Answer: B) Without translation

80. A linear transformation is represented by:

- A) $p' = Ap$
- B) $p' = Ap + b$

C) $p' = A + b$

D) $p' = A - b$

☒ Correct Answer: A) $p' = Ap$

81. The limitation of linear transformation is that it:

A) Cannot rotate

B) Cannot scale

C) Cannot translate because origin is anchored

D) Cannot shear

☒ Correct Answer: C) Cannot translate because origin is anchored

82. Homogeneous coordinates add which value to (x, y)?

A) 0

B) 1

C) -1

D) 2

☒ Correct Answer: B) 1

83. Homogeneous coordinates turn (x, y) into:

A) (x, y, 0)

B) (x, y, 1)

C) (x, y, -1)

D) (x, y, 2)

☒ Correct Answer: B) (x, y, 1)

84. Homogeneous coordinates allow:

A) Translation using matrix multiplication

B) Only rotation

C) Only scaling

D) Only reflection

☒ Correct Answer: A) Translation using matrix multiplication

85. A projective transformation is also called:

A) Homography

B) Histogram

C) Convolution

D) Thresholding

☒ Correct Answer: A) Homography

86. A projective transformation uses a:

A) 2×2 matrix

B) 2×3 matrix

C) 3×3 matrix

D) 4×4 matrix

✓ Correct Answer: C) 3×3 matrix

87. Projective transformations allow:

- A) Parallel lines to remain parallel
- B) Parallel lines to converge
- C) Only rotation
- D) Only scaling

✓ Correct Answer: B) Parallel lines to converge

88. In affine transformation, the bottom row of the matrix is:

- A) [1, 1, 1]
- B) [0, 0, 1]
- C) [0, 1, 0]
- D) [1, 0, 0]

✓ Correct Answer: B) [0, 0, 1]

89. The denominator effect in projective transformation creates:

- A) Depth/perspective
- B) Color change
- C) Blur
- D) Noise

✓ Correct Answer: A) Depth/perspective

90. Projective transformation can model:

- A) Converging lines and camera viewpoints
- B) Only 2D rotation
- C) Only grayscale conversion
- D) Only thresholding

✓ Correct Answer: A) Converging lines and camera viewpoints

91. Planar alignment uses which transformation?

- A) Linear
- B) Affine
- C) Projective
- D) Photometric

✓ Correct Answer: C) Projective

92. The key difference between affine and projective is:

- A) Affine preserves parallelism; projective does not
- B) Affine changes color; projective does not
- C) Affine is 3D; projective is 2D
- D) Affine uses 4×4; projective uses 2×3

✓ Correct Answer: A) Affine preserves parallelism; projective does not

93. Linear transformations can achieve:

- A) Scaling, rotation, reflection
- B) Translation only
- C) Perspective only
- D) Histogram equalization

✓ Correct Answer: A) Scaling, rotation, reflection

94. In a homogeneous 3×3 matrix, translation values are placed in:

- A) The rightmost column
- B) The bottom row
- C) The diagonal
- D) The first row

✓ Correct Answer: A) The rightmost column

95. Composition of transformations is powerful because:

- A) Multiple transformations can be combined into one master matrix
- B) It removes all noise
- C) It converts RGB to BGR
- D) It increases resolution

✓ Correct Answer: A) Multiple transformations can be combined into one master matrix

96. In OpenCV, positive rotation angle in `getRotationMatrix2D` is:

- A) Clockwise
- B) Counter-clockwise
- C) Always 0
- D) Always 180

✓ Correct Answer: B) Counter-clockwise

97. Which of the following is a geometric transformation?

- A) Brightness adjustment
- B) Rotation
- C) Gamma correction
- D) Color conversion

✓ Correct Answer: B) Rotation

98. Which of the following is a photometric transformation?

- A) Translation
- B) Scaling
- C) Contrast enhancement
- D) Shearing

✓ Correct Answer: C) Contrast enhancement

99. The image coordinate system is important because:

- A) It affects how transformations like rotation and cropping are applied

- B) It changes pixel colors
- C) It compresses images
- D) It equalizes histograms

☒ Correct Answer: A) It affects how transformations like rotation and cropping are applied

100. A 2D affine transformation can be represented as:

- A) 2×2 only
- B) 2×3 or 3×3 homogeneous
- C) 4×4 only
- D) 1×2 only

☒ Correct Answer: B) 2×3 or 3×3 homogeneous